| | | Target engagement | Dose | Efficacy | Safety |
|---|---|---|---|---|---|
| #15 | **hepatitis C**<br>HCV NS5A inhibitors (10) | **2011** HCV subgenomic replicon in Huh-7 cells, genotype 1a/1b and patient-derived NS5A hybrids | **2015** HCV subgenomic replicon in Huh-7 cells, genotype 1a/1b and patient-derived NS5A hybrids | **2020** HCV viral-kinetic model of on-treatment HCV RNA decline, later run in real time to set DAA duration | **2015** Rodent, dog and monkey tox<br>Rat/monkey adrenal signal absent in patients |
| | **hepatitis C**<br>NS5B non-nucleoside inhibitors (10) | **2014** HCV genotype 1 subgenomic replicon (Huh-7), with and without human serum | **2014** Rat and dog repeat-dose toxicology (dasabuvir) | **2018** Human-hepatocyte chimeric mouse, patient HCV, 4-week DAAs<br>Replicon potent; filibuvir no SVR | **2014** Rat and dog repeat-dose toxicology (dasabuvir) |
| | **hepatitis C**<br>HCV NS5B (nucleotide) inhibitors (7) | **2012** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons) | **2013** Sofosbuvir rat/dog/mouse tox | **2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons) | **2013** IND-era in-vitro mitochondri…<br>HepG2/CEM assay cleared BMS-986094; heart failure |
| #16 | **diabetic kidney disease**<br>NR3C2 antagonists (3) | **2021** Rat urinary Na+/K+ natriuresis under MR-agonist challenge | **2021** Phase 2b population PK/exposure-response model of UACR, serum potassium and eGFR | **2020** Uninephrectomised DOCA-salt rat | **2021** Rat autoradiography and equinatriuretic DOCA-salt<br>No rat hyperkalaemia; DKD had it |
| #17 | **acute myeloid leukaemia**<br>FLT3 inhibitors (4) | **2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3) | **2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3) | **2012** Ba/F3 FLT3-ITD resistance screen<br>Combination: no survival gain | **2018** Rat and dog toxicology and safety pharmacology<br>PRES and QT missed by tox species |
| | **acute myeloid leukaemia**<br>CD33-directed depleters (3) | **2021** AML blasts + NK cells ex vivo<br>NK activation unchanged despite CD33 saturation | **2018** CD33+ AML xenografts (mice)<br>Phase 3 regimen failed, program abandoned | **2025** Cord-blood CD33-CAR-NK: Molm13 xenografts, HSC colonies<br>No single-agent responses in AML | **2025** Cord-blood CD33-CAR-NK: Molm13 xenografts, HSC colonies |
| #18 | **malignant pancreatic neoplasm**<br>MAP2K1/2 inhibitors (3) | **2012** Tumour xenografts, pERK by IHC (trametinib) | **2013** Pancreatic cell-line and PDX<br>Target reached, no benefit, programme dropped | **2018** RAS/BRAF lines, pimasertib<br>KRAS status did not pick pimasertib responders | **2013** Rat, dog, rabbit, mouse safety<br>Animals missed ocular harm, LVEF late |
| #19 | **MASH**<br>NR1H4 agonists (4) | **2019** Rodent FXR target-gene PD (tropifexor) | **2019** STAM and AMLN-diet NASH mice (tropifexor) | **2021** STAM and AMLN-diet NASH mice (tropifexor)<br>Outcome trial negative | **2024** High-fat-diet mouse ALT/cholesterol screen<br>Cilofexor: no lipid change |
| | **MASH**<br>FGF21 analogues (3) | **2020** Obese/diabetic cynomolgus monkeys | **2023** Pegozafermin population PK/PD exposure-response model<br>Reached, fibrosis failed, dropped | **2023** Pegozafermin population PK/PD exposure-response model<br>Pegbelfermin failed on fibrosis | **2020** FGF21 rs838133 and truncating-variant carriers (UK…<br>No mouse bone loss at 16 weeks |
| #20 | **prostate cancer**<br>AR antagonists (5) | **2013** LNCaP/AR castrate xenografts | **2013** LNCaP/AR castrate xenografts | **2016** LNCaP/AR castrate xenografts | **2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide) |
| | **prostate cancer**<br>PARP1/2 inhibitors (5) | **2009** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone | **2015** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone<br>Exposure OK; HRR-negative futile | **2023** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone<br>HRR-negative arm stopped, futile | **2023** Rat, dog repeat-dose tox of olaparib, niraparib, talazoparib<br>Anaemia, neutropenia unforeseen |
| | **prostate cancer**<br>CYP17A1/AR inhibitors (3) | **2018** VCaP mice, cyno/rat steroids | **2018** VCaP mice, cyno/rat steroids<br>Exposure plateaued; PK stopped development | **2015** LAPC4 androgen-dependent xenografts in SCID mice, galeterone<br>1/17 PSA50 after enzalutamide | **2018** VCaP mice, cyno/rat steroids<br>CNS tox closed prostate phase 2 |
| | **prostate cancer**<br>EZH2 inhibitors (3) | **2019** Human CRPC cell lines and xenografts (EZH2 inhibitor + AR blockade) | **2026** Human CRPC cell lines and xenografts (EZH2 inhibitor + AR blockade) | **2026** Human CRPC cell lines and xenografts (EZH2 inhibitor + AR blockade) | **2026** Rat and monkey repeat-dose toxicology (tazemetostat) |
| | **prostate cancer**<br>FOLH1-directed depleters (3) | **2019** Docetaxel-refractory PSMA-high CRPC xenografts dosed with PSMA ADC | **2011** Docetaxel-refractory PSMA-high CRPC xenografts dosed with PSMA ADC | **2022** PSMA-positive vs PSMA-negative prostate tumour-bearing mice | **2022** GLP single-dose rat, minipig and 4-week rat tox, cold PSMA-617<br>Low rodent PSMA hid xerostomia |
| #21 | **follicular lymphoma**<br>CD19-directed depleters (3) | **2011** Syngeneic anti-murine-CD19 CAR T cells vs 38c13 CD19+ lymphoma in C3H mice | **2015** Raji CD19+ Burkitt xenograft in NSG mice, FMC63-4-1BB CAR T cells (liso-cel lineage) | **2024** Raji CD19+ Burkitt xenograft in NSG mice, FMC63-4-1BB CAR T cells (liso-cel lineage) | **2023** Humanized NSG mice, high B-ALL burden, CD19 CAR T cells<br>CRS/neurotox despite lenzilumab |
| | **follicular lymphoma**<br>PIK3CD inhibitors (3) | **2011** Patient B-cell tumour samples and B-cell lines ex vivo | **2024** Healthy-volunteer whole-blood basophil activation test, ex-vivo PI3Kδ PD assay for ME-401 | **2014** Patient B-cell tumour samples and B-cell lines ex vivo | **2014** Rat and dog repeat-dose toxicology |
| #22 | **ovarian cancer**<br>PARP1/2 inhibitors (6) | **2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib) | **2021** NOVA exposure–safety model (niraparib) | **2016** BRCA-deficient cancer cells and xenografts (niraparib) | **2017** Dog cardiovascular safety pharmacology (niraparib) |
| #23 | **rheumatoid arthritis**<br>BTK inhibitors (8) | **2013** BTK occupancy probe in CIA mice | **2022** CIA occupancy-efficacy PK/PD<br>Target reached, no effect | **2019** Collagen-induced arthritis (rodent)<br>CIA-active drug failed Phase 2 | **2021** Human platelet aggregometry and humanized-mouse thrombus formation (acalabrutinib vs ibrutinib) |
| | **rheumatoid arthritis**<br>SYK inhibitors (3) | **2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) | **2014** Rat adjuvant- and collagen-induced arthritis (MK-8457)<br>Target hit, DAS28 flat, RA dropped | **2014** Rat adjuvant- and collagen-induced arthritis (MK-8457)<br>DAS28 no better than placebo | **2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib)<br>Serious infections missed, ended |
| | **rheumatoid arthritis**<br>anti-CSF2 antibodies (3) | **2018** Mouse GM-CSF-IRF4-CCL17 pathway in zymosan/antigen arthritis pain models | **2025** Cynomolgus 13-week SC and 26-week IV GLP tox, otilimab<br>Weekly OK; targeted-IR failed | **2022** Mouse arthritis, anti-GM-CSF<br>Failed in targeted-therapy-IR; met ACR20 elsewhere | **2025** Human anti-GM-CSF autoantibody alveolar proteinosis<br>No PAP, 2915 patients, 2.5 years |
| #24 | **multiple myeloma**<br>TNFRSF17-directed depleters (8) | **2016** BCMA+ myeloma xenografts in NSG mice, CAR-T | **2026** Belantamab PopPK + M-protein tumour-growth + keratopathy Markov simulation… | **2019** BCMA+ myeloma xenografts in mice, ADC | **2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin) |
| | **multiple myeloma**<br>PSMB5 inhibitors (3) | **2015** MM.1S myeloma xenografts in SCID mice, β5 PD, ixazomib | **2015** Ixazomib population PK, fixed 4 mg versus BSA-based dosing<br>Dosed to MTD, failed, abandoned | **2015** MM.1S myeloma xenografts in SCID mice, β5 PD, ixazomib<br>9% PR vs marked PDX activity | **2015** Ixazomib GLP tox, 6-month rat and 9-month dog repeat dose<br>Tolerated in dogs; GI tox ended it |
| #25 | **melanoma**<br>BRAF inhibitors (3) | **2010** BRAF V600E cell lines and xenografts | **2011** Colo205 BRAF V600E xenograft (exposure-response) | **2010** BRAF V600E vs wild-type cell-line panel (patient selection)<br>INTERIM: intermittent worse again | **2014** HRAS Q61L keratinocytes and DMBA/TPA mouse skin<br>FDA: cuSCC ~29%, animals blank |
| | **melanoma**<br>anti-PDCD1 antibodies (3) | **2021** Humanised-PD-1 C57BL/6 mouse with S180 homografts, 124I-toripalimab immuno-PET | **2019** Human T-cell PD-1 occupancy in vitro (saturation threshold) | **2026** Human PD-1 knock-in mouse bearing MC38 tumours | **2024** Monkey 4/26-week GLP tox<br>Endocrine and skin irAEs in melanoma cohort |
| #26 | **Crohn disease**<br>anti-IL23A antibodies (3) | **2018** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb) | **2022** Sponsor population-PK / exposure-response dose model from phase 2, risankizumab, mirikizumab | **2017** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb) | **2022** Monkey tox, risankizumab<br>Hepatotoxicity in Crohn's; monkeys silent |
| #27 | **open-angle glaucoma**<br>ROCK inhibitors (3) | **2018** Normotensive monkey eye, topical netarsudil 0.04% | **2014** Normotensive monkey eye, topical netarsudil 0.04% | **2014** Normotensive monkey eye, topical netarsudil 0.04%<br>Genotype did not predict response | **2017** Monkey 9-month topical ocular toxicology |
| Allergy | **allergic disease**<br>allergen (immunotherapy) (5) | **2019** Patient PBMC Fel d 1-specific CD4+ T cells (ex vivo), Cat-PAD | **2016** Allergen exposure chamber<br>Chamber regimen reached in field, failed | **2018** Mouse airway allergy, Fel d 1<br>Mouse protection, field loss to placebo | **2019** Allergen exposure chamber |
| | **allergic rhinitis**<br>PTGDR2 antagonists (4) | **2013** Human whole-blood PGD2 eosinophil shape-change assay | **2014** Asthmatic allergen challenge<br>Challenge regimen reached, failed, abandoned | **2017** OVA-sensitised mouse…<br>Mouse nasal-symptom relief absent in Phase 3 | **2017** Asthmatic allergen challenge |
| | **allergic rhinitis**<br>allergen (immunotherapy) (8) | **2018** Conjunctival test, gp-ASIT+ | **2019** Conjunctival test, gp-ASIT+ CPT dose reached, confirmatory trial missed | **2019** Conjunctival test, gp-ASIT+<br>Confirmatory field trial missed primary | **2019** Allergen exposure chamber |
| Autoimmune | **psoriasis**<br>JAK1 inhibitors (3) | **2018** Human whole-blood cytokine-induced STAT phosphorylation, JAK1 selectivity | **2022** Rat, monkey repeat-dose tox | **2018** Rat adjuvant-induced arthritis (cross-indication; no psoriasis model was run) | **2022** Rat, monkey repeat-dose tox<br>Dose-related platelet fall the animals missed |
| | **psoriasis**<br>PDE4 inhibitors (3) | **2020** Human Th17-driven ex vivo skin inflammation model, LEO 29102 cream 2.5 mg/g | **2011** Mouse phorbol ear edema<br>Regimen reached, Phase-3-design trial failed | **2019** In-vitro PDE4 enzyme and cellular assays, skin-delivery design<br>Imiquimod mouse yes, patients no | **2011** Rabbit primary skin irritation and mouse local lymph node assay (2% ointment) |
| | **psoriasis**<br>RORC inverse agonists (4) | **2021** Human whole-blood IL-17 secretion assay<br>No IL-17 change in plaques | **2021** Human whole-blood IL-17 secretion assay | **2022** IL-23 minicircle mouse, oral BI 730357 | **2015** Nonclinical toxicology species (not public) |
| | **psoriasis**<br>anti-IL17A antibodies (4) | **2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout | **2016** Cynomolgus repeat-dose toxicology | **2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout | **2016** People with inborn errors of IL-17 immunity |